Mind-wandering preserves word-level coupling but produces localized comprehension failure

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Abstract

Mind-wandering during reading creates a familiar paradox: the eyes continue across the page, yet little is retained. This is commonly explained as perceptual decoupling, in which attention withdraws from the text and word processing no longer guides the eyes. We asked whether the failure instead occurs later, after individual words have been processed but before they are integrated into a coherent representation of the passage.

Forty-four readers viewed connected articles while eye movements and electroencephalography were recorded, and afterwards identified the periods in which their minds had wandered. Sensitivity to word frequency, length and contextual predictability was preserved during these episodes across three separable decisions: whether to fixate a word, how long to remain, and whether to return. In each case it was statistically equivalent to on-task reading within ±10%. A fixation-locked occipitotemporal response indexing lexical processing was also preserved, to within 8%. The centroparietal response to contextual predictability could not be estimated reliably here, so the semantic channel remains open. The same measures detected attenuated word-to-eye coupling under an instructed shallow-reading goal in an independent sample, so their stability during spontaneous episodes was not a failure of measurement.

The most frequently cited marker of mindless reading, an increase in skipping, proved to be an artifact of how skipping is scored in continuous text. Under a scan-path criterion readers skipped 18.5% *fewer* words while mind-wandering, and read the remainder more effortfully.

What changed was not sensitivity to individual words but the outcome of reading. Comprehension deteriorated specifically for the material covered by the reported lapse: accuracy fell from 0.68 to 0.41, exceeding the effect of all 1000 size-matched regions elsewhere on the same page. Where attention had wandered predicted what was later lost; where the eyes had looked did not.

Mind-wandering during sustained reading therefore need not disconnect the reader from the words themselves. The breakdown appears to occur downstream of word identification, in the processes that turn successive words into an enduring representation of the text.

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1 Introduction

Everyone who reads has had the experience of reaching the bottom of a page and realizing that nothing has gone in. The eyes moved, the lines were traversed, and yet no meaning accumulated. Laboratory work has turned this ordinary failure into a tractable phenomenon: readers probed during self-paced reading report task-unrelated thought on a sizeable minority of trials, and those reports are followed by measurable changes in the eye-movement record and in comprehension (Smallwood and Schooler, 2006; Schooler et al., 2011; Smallwood and Schooler, 2015).

The standard explanation is perceptual decoupling. On this account attention withdraws from the external input during mind-wandering, so the products of word identification no longer reach the systems that program eye movements (Smallwood, 2011; Schooler et al., 2011). The account makes a sharp and testable prediction. If lexical processing is what normally makes the eyes behave differently for a rare word than for a common one, then the difference should shrink when the mind is elsewhere. Reichle et al. (2010) reported exactly this pattern in a self-caught paradigm: fixations before self-reported mindless reading were longer and less sensitive to word frequency and length. Schad et al. (2012) showed that the disruption is graded, with deeper lapses producing larger changes. A recent meta-analysis of eye-movement markers concluded that the most consistent signatures are an increase in skipping and a reduction in fixations per word, and that reduced sensitivity to lexical properties is diagnostic of the deeper form of mindless reading (Mézière et al., 2026).

Two features of that evidence base leave the central claim less settled than it appears.

First, the claim is about coupling, and coupling has been measured almost entirely in one place. Reading is not a single decision. Models that differ in almost every other respect agree that the eyes make at least three separable choices about each word: whether to fixate it at all, how long to remain once fixated, and whether to come back (Reichle et al., 2003; Engbert et al., 2005; Rayner, 2009). These decisions draw on lexical information at different points and with different latencies. Skipping is decided before the word is fixated, on parafoveal information, and is strongly driven by length and frequency (Rayner, 1998; Kliegl et al., 2004). Dwell time reflects processing of the fixated word. Regressions are a repair mechanism deployed after difficulty has been encountered. A withdrawal of attention that spared one of these and abolished another would look very different from one that scaled them all (Fig. 1b). Fixation duration alone cannot distinguish these possibilities.

Second, and more practically, fixation duration can only be measured on words that were fixated. Whether a word is fixated is itself the most lexically driven decision the reader makes, and it changes during mind-wandering. Conditioning a coupling estimate on a variable that the manipulation moves is a familiar source of bias. It matters here because the measurement of skipping in continuous multi-page reading is not straightforward. A word can fail to receive a fixation because the reader stepped over it in a single saccade, which is skipping in the sense the models mean, or because gaze and text were transiently out of register, which is not.

A third gap is neural. Simultaneous eye tracking and electroencephalography now makes it possible to read out word processing directly, without inferring it from behavior. Fixation-related potentials in natural reading carry a graded occipitotemporal response to word frequency from roughly 150 ms and a centroparietal N400 that scales with contextual surprisal (Dimigen et al., 2011; Kutas and Federmeier, 2011; Frank et al., 2015). If mind-wandering gates the input, these responses should shrink. Almost all co-registered reading work has been conducted on-task, and almost all mind-wandering reading work has been behavioral. We are not aware of a well-powered test at the intersection.

We addressed all three gaps in a single design, and then asked what the reader actually loses, using the page-level comprehension questions that the dataset records.

In the primary dataset, 44 readers read five connected Wikipedia articles at their own pace while 64-channel electroencephalography and binocular eye movements were recorded, and reported the spans during which their minds had wandered. This yielded 404,557 first-pass fixations, of which 7.2% fell inside a reported span. We measured word sensitivity at all three levels of eye-movement control and in two fixation-locked neural responses, and we treated the null as something to be supported rather than merely observed, using equivalence tests and Bayes factors throughout. Because a null is only as good as the sensitivity of the instrument that produced it, we included a positive control: an independent sample of 12 readers who read under an instructed shallow-search goal as well as for comprehension (Hollenstein et al., 2018a). If our measures can detect a change in coupling when one is present, then their failure to detect one during mind-wandering is informative.

What we found is a single dissociation with two halves. During mind-wandering the eyes go on doing word-sensitive reading: every measure of how strongly word properties govern the reading system was preserved, in behavior and in the fixation-locked brain response. What fails is what is built from those words: comprehension collapses, and it collapses specifically on the material the reader was absent for, while the record of where the eyes went carries no such information. The rest of the paper establishes those two halves. It also shows that the same measures do register a change in coupling when one is genuinely present, and identifies a measurement artefact that explains part of why the field has read the first half differently.

2 Results

The order below follows that dissociation. The first two subsections establish the instrument and show that the reported spans behave like mind-wandering. The next four establish its first half. Word sensitivity is preserved at all three levels of eye-movement control and in the lexical brain response. The canonical contrary marker does not survive a scan-path definition of skipping. The corrected behavioral profile is one of more effort rather than less. An instructed shallow goal moves the same measures that mind-wandering leaves alone. The last two subsections establish the second half. What does change is global and word-independent, and comprehension fails on the specific content the reader was absent for.

2.1 Reading couples the eyes and the brain to word properties

We first established that the measures respond to word properties in the expected way while readers are on task (Fig. 2).

Fixation durations tracked word properties in a mixed model with by-reader random slopes: log fixation duration decreased with word frequency ($\beta = -0.0178$, SE = 0.0025, $p = 4.6 \times 10^{-13}$) and increased with GPT-2 contextual surprisal ($\beta = +0.0159$, SE = 0.0016, $p = 1.9 \times 10^{-23}$).

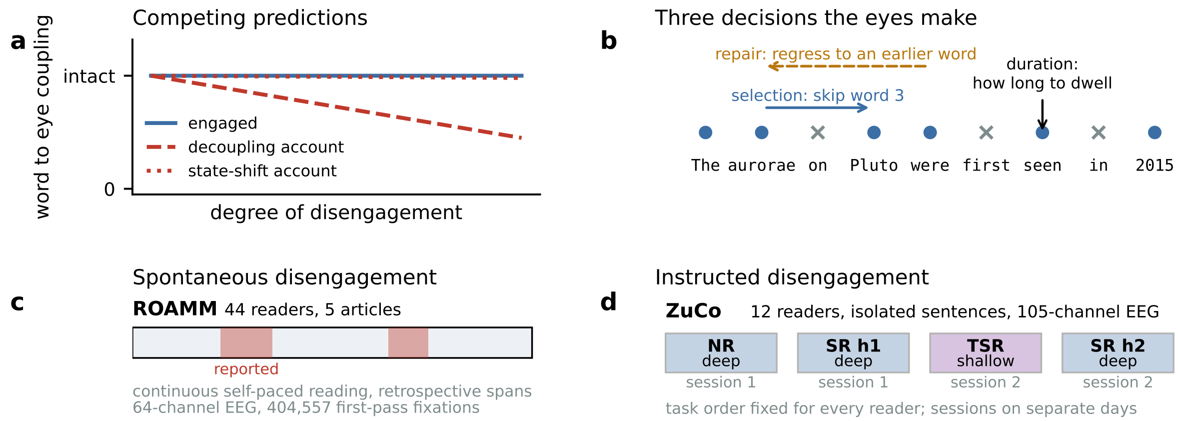

Figure 1: Question and design. (a) Perceptual decoupling predicts that the influence of word properties on eye movements weakens as disengagement deepens; an account in which disengagement is a change of global state predicts that it does not. (b) The eyes make at least three separable decisions about each word. Circles mark fixated words and crosses mark words that were stepped over. (c) The primary sample. Readers read connected articles at their own pace and afterwards marked the spans during which their minds had wandered. (d) The control sample. Readers read isolated sentences under a deep or a shallow goal. Task order was the same for every reader and the two encyclopedic tasks fell in different sessions, which we exploit in Section 2.6.

Word properties also predicted the other two decisions. Using Somers' D, a rank-based association measure that is insensitive to base rates, frequent and short words were more likely to be skipped ($D = +0.50$ for frequency, $D = -0.56$ for length) and less likely to be surprising ($D = -0.38$). Corrective returns to a skipped word were more likely when the skipped word was rare, long or surprising (frequency $D = -0.25$, $p = 3.4 \times 10^{-11}$; length $D = +0.25$, $p = 1.7 \times 10^{-13}$; surprisal $D = +0.21$, $p = 9.5 \times 10^{-14}$), which is what a repair mechanism should do.

Fixation-locked potentials showed the two canonical effects after overlap correction by regression ERP (Smith and Kutas, 2015; Ehinger and Dimigen, 2019). The deconvolved frequency kernel produced an occipitotemporal response between 150 and 290 ms (+0.149 µV per SD, $t_{43} = 8.06$, $p = 3.9 \times 10^{-10}$), and the surprisal kernel produced a centroparietal negativity between 300 and 450 ms (−0.038 µV per SD, $t_{43} = -5.31$, $p = 3.6 \times 10^{-6}$). The frequency response replicated in the independent 12-reader sample recorded on different hardware in a different laboratory (+0.139 µV per SD, sign-flip cluster $p = 0.040$). The surprisal N400 did not replicate there ($p = 0.82$; cluster $p = 0.22$), which is expected given isolated single sentences and a fifth of the sample size, so the semantic channel is carried by the primary dataset alone.

2.2 The reported spans behave like mind-wandering

Because the primary sample uses retrospective span marking rather than probes, we checked that the labels track the canonical behavioral signature before treating any null based on them as informative. All 44 readers marked at least one span, per-reader rates ranged from 0.2% to 24.2% of fixations, and fixations inside a reported span were 5.0% longer than the same reader's on-task fixations (95% CI [3.3, 6.6], $p = 7.9 \times 10^{-7}$,

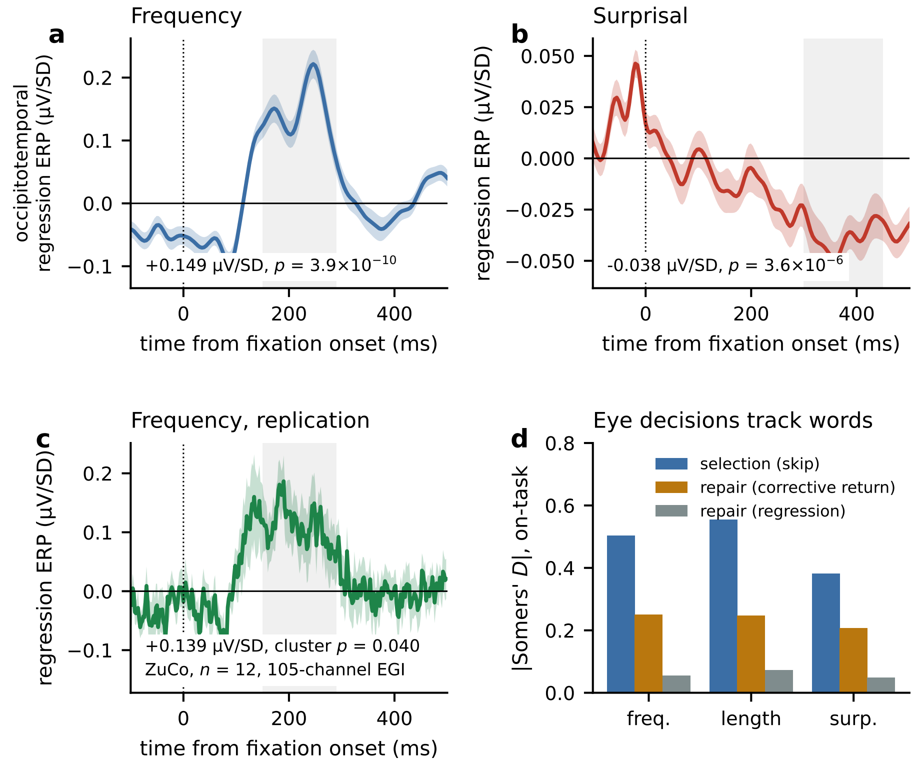

Figure 2: The measures respond to word properties during engaged reading. (a) Overlap-corrected regression-ERP kernel for word frequency at occipitotemporal sensors. (b) Kernel for contextual surprisal at centroparietal sensors. Shading is the standard error across 44 readers; gray bands mark the a priori analysis windows. (c) The frequency kernel in an independent 12-reader sample recorded with different hardware. (d) Rank-based association between word properties and each of the three eye-movement decisions during on-task reading.

39 of 44 readers). The labels reproduce the effect that mindless reading is best known for.

2.3 Mind-wandering leaves selection, repair and duration intact

We then asked whether any of these sensitivities changed during reported mind-wandering (Fig. 3). This is the first half of the dissociation, and it is a null we treat as a quantity to be bounded rather than a test to be passed.

Selection. A word counted as skipped only if the reader's scan path stepped over it in a single forward saccade spanning at most four intervening words. Words lying inside larger positional gaps were excluded rather than scored. Those gaps are almost entirely return sweeps between text lines (97% of them, see below), and the words inside them are passed over for reasons of display layout rather than of lexical access. This yielded 332,016 words, of which 21,633 fell inside a reported mind-wandering span, from 36 readers with at least 150 words in each state. The influence of word properties on skipping was unchanged. Retention of the on-task association was 100.5% for frequency (mind-wandering minus on-task $\Delta D = +0.0025$, 95% CI $[-0.0167, +0.0214]$, $p = 0.80$), 101.8% for length ($\Delta D = -0.0100$, $[-0.0283, +0.0099]$, $p = 0.32$) and

103.1% for surprisal ($\Delta D = -0.0118$, $[-0.0326, +0.0099]$, $p = 0.30$). Expressed relative to the on-task effect, the frequency interval runs from −3.3% to +4.2%. All three were statistically equivalent within ±10% of the on-task effect (TOST $p = 1.1 \times 10^{-5}$, 2.6×10^{-5} and 0.011; $\mathrm{BF}_{01} = 5.4$, 3.5 and 3.3). We set the smallest effect of interest from the literature rather than from these data. Reichle et al. (2010) report frequency and length effects that are essentially abolished before a self-caught lapse, Schad et al. (2012) report attenuations on the order of a third to a half at their deeper level, and the instructed-goal contrast reported below attenuates the same slope by 34 to 44%. A bound of 20% is therefore already well inside the range the decoupling account predicts, and 10% is conservative by a further factor of two.

These nulls survived every control we ran. They were stable across skip-definition thresholds from one to ten intervening words, unchanged when restricted to spans the reader marked as fully off-task, present within each tertile of position through a run, and stable when any one article was dropped. A relocation control, in which each reader's mind-wandering spans were moved to random on-task positions of matched length and re-analyzed 200 times, reproduced the observed values ($p_{\mathrm{perm}} = 0.81$, 0.28 and 0.31), so even the small numerical trends are not specific to mind-wandering. For context on sensitivity, the width of the frequency interval means that an attenuation of the size the instructed-goal condition produces below, roughly a third, would have been detected here with a very large margin.

Repair. Refixation selectivity was retained at 96% to 99% (35 readers, all $p > 0.5$) and the tendency of regressions to target difficult words was retained at 74% for frequency ($p = 0.25$), 73% for length ($p = 0.056$) and 100% for surprisal ($p = 0.99$; 38 readers). The length effect is the only hint of attenuation anywhere in the battery and it does not survive correction across the three properties. Corrective-return selectivity could be estimated in only 16 readers, because it requires enough single-skip events in both states within a reader, and we therefore treat it as uninformative rather than as evidence either way. The corrective-return *rate*, which uses all 44 readers, is reported below.

Duration. Across words, the mind-wandering slope was slightly larger than the on-task slope, with signed retention of 114.6% for frequency (paired $p = 0.001$) and 111.6% for length (paired $p = 0.007$). We do not interpret this as sharpened lexical control, for two reasons. First, length is a visual property, and it was enhanced as much as frequency, which is the signature of a global rescaling of log slopes rather than of anything lexically specific. Second, the effect is not identified across words, because the words a reader happens to encounter while mind-wandering are not a random sample. Since each corpus token was read by up to 44 readers, and 7,264 of 10,183 tokens (71.3%) were read both on-task and during mind-wandering by different readers, mind-wandering status varies within token. A two-way fixed-effects model absorbing token identity and reader identity, with reader-clustered standard errors, roughly halved the estimate and removed its significance (frequency $\beta = -0.0109$, SE = 0.0070, $p = 0.13$; surprisal $\beta = -0.0084$, $p = 0.18$; length $\beta = +0.0087$, $p = 0.27$; $n = 234{,}800$ word observations). The additive component was large and highly reliable in the same model ($\beta = +0.103$, SE = 0.011, $p = 1.6 \times 10^{-11}$). Mind-wandering moved the intercept, not the slope.

Beyond three chosen measures. Each test above fixes a decision and a word property in advance, which leaves open the possibility that we chose the measures that do not move. We therefore replaced the choice with a model. For every fixation a network predicted which of the 41 words within twenty positions the reader would move to next, and how long the current fixation would last, from two blocks of information held apart: page geometry and reading history on one side, the word properties of the candidates on the other. Networks of this kind now predict reading scan paths well (Deng et al., 2023; Bolliger et al., 2025) and have been used to model task effects (Hahn and Keller, 2023); we use one only as a measuring device. It was fitted on on-task fixations alone and evaluated in cells where both the reader and the article were held out. The read-out is how much the word-property block improves held-out prediction, in bits. This is a different instrument from fitting a mechanistic model of eye-movement control to an instructed mindless condition and reading decoupling off a parameter (Nuthmann and Engbert, 2009): it commits to no mechanism, and it returns a bound.

On-task, knowing the words was worth 0.144 bits per movement (all 44 readers positive, $p = 1.8 \times 10^{-25}$) against a geometry-only baseline of 2.70 bits out of the 5.36 available from the candidate set, so the probability the model places on the word actually fixated next rises by about a tenth. Permuting which word occupies which position within a page, leaving layout, timing and the reader's own movements untouched, removes the gain entirely (−0.013 bits, 4 of 44 readers). What the model uses is word identity and not geometry under another name.

During mind-wandering this measure retained 0.911 of its on-task value (95% CI [0.822, 0.995]), with a change of 12% detectable at 80% power; the duration head retained 1.071 [0.737, 1.430] and is far less informative (Table 1). The 9% shortfall in target choice is the only contrast in this study whose interval excludes zero, and we do not present it as established. It survives every geometry control we can absorb, including launch line, word identity and reader by page (β from −0.015 to −0.013, $p = 0.019$ at the strictest rung), and that same ladder recovers the known additive lengthening of fixations (+4.2% falling to +2.9%, all $p < 10^{-9}$). It does not survive an episode-preserving permutation in which each reader's spans are replaced by another reader's at the same positions in the same text ($z = -1.84$, $p = 0.067$, 2000 draws), and the most outlier-resistant estimator returns 0.942 [0.878, 1.005]. The shortfall is also uniform across kinds of movement: in bits it is −0.012 for forward saccades and −0.011 for regressions and refixations, and the interaction between state and movement kind is null ($p = 0.94$). Section S10 reports eleven further exclusions, across which retention stays between 0.88 and 0.97.

Brain. The two neural channels differ sharply in what they can support, and we separate them.

The occipitotemporal frequency response was preserved. In the overlap-corrected model the frequency by mind-wandering interaction was +0.024 µV ($p = 0.30$, $n = 44$), which is +16.0% of the on-task effect with a 95% interval of [−12.5, +45.9]%. Because the decoupling account predicts a reduction, the informative bound is one-sided: attenuations greater than 8% are excluded ($BF_{01} = 3.6$). This response therefore behaves like the behavioral measures.

The centroparietal surprisal response could not be tested usefully. The interaction was +0.005 µV ($p = 0.79$), which is −13.4% of the on-task effect, but with a 95% interval of [−112.2, +81.6]%. The

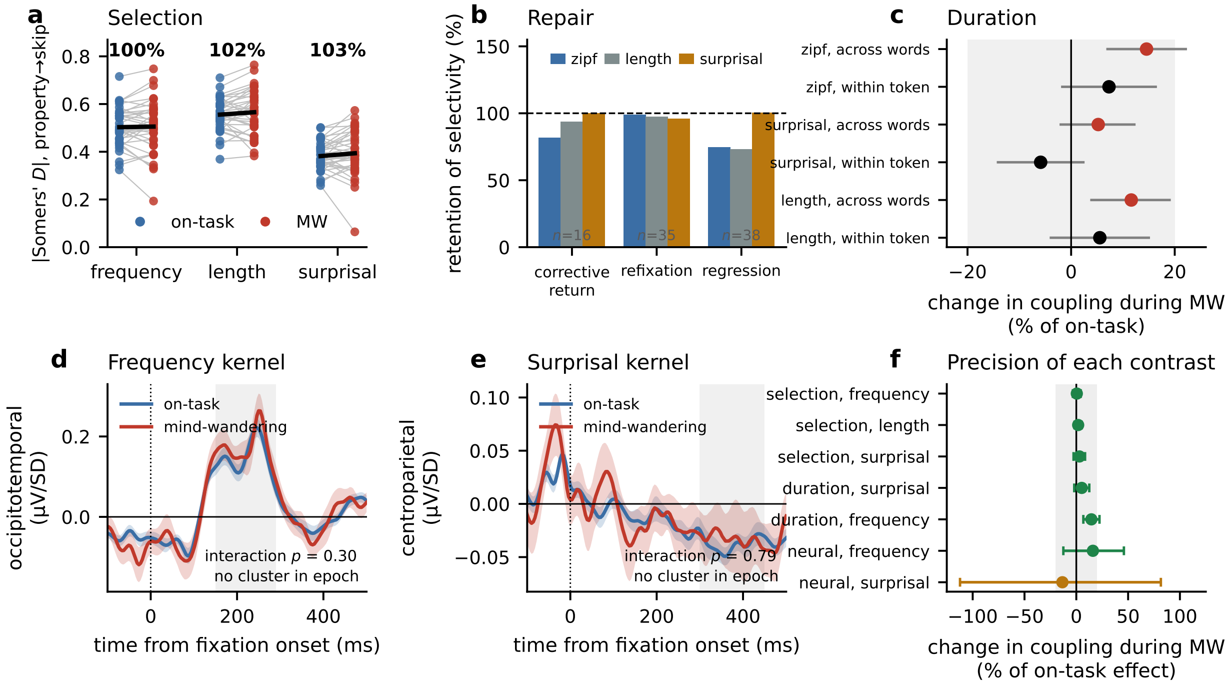

Figure 3: Mind-wandering spares word sensitivity at every level measured. (a) Rank-based association between word properties and skipping, per reader, on-task versus mind-wandering; black lines join condition means, percentages give retention. (b) Retention of selectivity for the three repair behaviors. (c) Change in the fixation-duration slope, estimated across words (colored) and within token instance with reader and token fixed effects (black). Bars are 95% intervals; the shaded band is the ±20% equivalence region. (d, e) Overlap-corrected kernels by state. (f) Change in coupling for every contrast, with 95% bootstrap intervals. Orange marks the surprisal response, whose interval is too wide to constrain the question at all; the other contrasts each exclude attenuations of more than a few percent.

smallest change this design could have detected at 80% power is 144% of the on-task effect, so the test cannot distinguish a preserved N400 from an abolished one. We report it as uninformative rather than as evidence of preservation. The limit is the signal-to-noise ratio of the surprisal response itself, which is reliable at the group level ($p = 3.6 \times 10^{-6}$) but small in absolute terms ($-0.038\,\mu$V per SD), so a 30% modulation of it lies far below what 44 readers can resolve. A cluster-based permutation test over the whole epoch and all sensors returned no cluster for either interaction (min $p = 1.0$), which is consistent with both conclusions and decisive for neither.

The additive pattern seen in behavior appeared in the neural data as well: a uniform occipitotemporal amplitude offset of $+0.087\,\mu$V ($p = 3.4 \times 10^{-5}$) with the slope unchanged. Split-half reliability of the fixation-related potential was identical in the two states when trial counts were matched (0.888 versus 0.889, $p = 0.91$), so neither neural result reflects noisier data during mind-wandering.

Table 1 makes the asymmetry explicit. The three selection rows are the tightest in the study: each excludes attenuations above a few percent and could have detected a change of about 5 to 8%. The duration rows are comparable, except that the frequency contrast is displaced in the direction of stronger coupling, which the within-token model in Fig. 3c attributes largely to which words readers happened to encounter. The neural frequency row is looser but still useful, excluding attenuations above 8%. The neural surprisal

Table 1: Every coupling contrast, with what it can and cannot rule out. Δ is the change during mind-wandering as a percentage of the on-task effect, positive meaning stronger coupling. "Attenuation excluded" is the one-sided 95% bound, that is, the smallest reduction inconsistent with the data. The last column is the smallest change the contrast could have detected at 80% power. Neural rows use the overlap-corrected kernels from all 44 readers. The two omnibus rows are model-based and place no restriction on which word properties may be used; they are the only rows whose predictor is not chosen in advance.

Level	Predictor	Readers	Δ (% of on-task)	Attenuation excluded	Detectable at 80%
Selection	frequency	36	+0.5 [−3.3, +4.2]	> 3%	6%
Selection	length	36	+1.8 [−1.7, +5.2]	> 1%	5%
Selection	surprisal	36	+3.1 [−2.7, +8.5]	> 2%	8%
Duration	surprisal	38	+5.2 [−2.1, +12.4]	> 1%	11%
Duration	frequency	38	+14.6 [+6.7, +22.5]	none	12%
Omnibus	target choice	43	−8.9 [−17.8, −0.5]	> 16%	12%
Omnibus	duration	43	+7.1 [−26.3, +43.0]	> 21%	49%
Neural	frequency	44	+16.0 [−12.5, +45.9]	> 8%	43%
Neural	surprisal	44	−13.4 [−112.2, +81.6]	> 97%	144%

row excludes nothing: it could not have detected the complete disappearance of the effect, and no conclusion should be drawn from it in either direction. The two omnibus rows sit between these extremes and carry a different kind of weight, because nothing about them was chosen: they bound the attenuation of everything a high-capacity model can extract from the text at 16 and 21%. Both lie inside the 20% threshold set above from the literature, and far inside the third to one half that Schad et al. (2012) report at their deeper level.

2.4 Reconciling this with the reported increase in skipping

One result stands against the first half of the dissociation, and it is the most consistent behavioral marker in the meta-analytic literature: readers skip more words when their minds wander (Mézière et al., 2026). Our data reproduce it, and then overturn it (Fig. 4).

Scoring a word as skipped whenever it received no first-pass fixation anywhere in the reading span gives a large effect in the expected direction: skipping rises from 0.439 to 0.648 ($\Delta = +0.168$, 95% CI $[0.131, 0.206]$, $p = 7.4 \times 10^{-11}$, 41 of 44 readers). Requiring instead that the scan path demonstrably stepped over the word reverses the sign. With a threshold of at most four intervening words, skipping falls from 0.251 to 0.226 ($\Delta = -0.042$, $[-0.057, -0.028]$, $p = 1.8 \times 10^{-6}$, 38 of 44 readers). The reversal holds at every threshold from one to ten.

The two definitions differ over words that lie inside large positional gaps in the scan path, so what those gaps are matters. They are return sweeps. Clustering the bounding-box tops of the stimulus words into text lines, steps spanning more than four words cross a line boundary in 96.6% of on-task cases and 97.7% of mind-wandering cases, against 4.5% and 4.7% of steps between adjacent words (Fig. 4c). The words inside them are the last words of one line and the first words of the next, which readers pass over for reasons of layout rather than of lexical access, and which reading research conventionally treats separately. The threshold is therefore not a free parameter; it recovers a category defined by the display.

Nor are the gaps periods in which gaze leaves the text. Within a page, the fraction of time not spent on a mapped word was the same in the two states (0.166 versus 0.169, $p = 0.49$), and the interval between successive fixations was indistinguishable across states within every gap size (medians of 26 to 43 ms throughout). We note that the released derivative retains only fixations that could be mapped to a word. This is therefore an argument from elapsed time rather than a direct observation of where gaze went. An unobserved off-text fixation would have shown up as a long interval, and none does.

What differs between the states is structural. During mind-wandering 74% of stepped-over words lie inside large same-page forward jumps against 48% on-task, while 29% of on-task stepped-over words come from page transitions against under 1% during mind-wandering. Because reported spans rarely straddle a page change, the naive measure is assembling the two states from different kinds of gap.

The result does not depend on the exclusion. Excluding words is a state-dependent operation here, and the excluded words are not identical across states: relative to on-task, the words excluded during mind-wandering were slightly more frequent ($\Delta = +0.16$ Zipf, $p = 1.7 \times 10^{-4}$), shorter ($\Delta = -0.29$ characters, $p = 1.7 \times 10^{-6}$) and less surprising ($\Delta = -0.35$ bits, $p = 0.023$). We therefore repeated the primary analysis in two ways that do not rely on the gap criterion. Using a purely layout-based rule, in which a word is analyzable if the reader fixated at least one word on the same text line both before and after it, skipping still fell during mind-wandering (0.268 to 0.252, $\Delta = -0.035$, $p = 1.1 \times 10^{-5}$) and selectivity was still preserved (retention 96.9%, 97.6% and 99.1%; all $p > 0.12$; Fig. 4e). Matching the exclusion rate across states within reader gave the same answer for selectivity (99.4%, 98.6% and 101.0%; all $p > 0.4$). That procedure distorts the composition of the retained set, so we do not read a rate contrast from it.

2.5 Reading becomes more effortful, not more cursory

The corrected picture of what mind-wandering does to the eyes is coherent and is the opposite of skimming. Relative to on-task reading by the same readers, gaze durations were 14.6% longer (95% CI [10.3, 19.6], $p = 2.2 \times 10^{-7}$, 40 of 44 readers), regressions were 18.1% more frequent ($p = 1.3 \times 10^{-4}$, 39 of 44 readers), immediate refixations were 13.9% more frequent ($p = 0.006$, 34 of 44), corrective returns to skipped words were 12.6% more frequent ($p = 0.13$), and skipping was 18.5% less frequent ($p = 1.8 \times 10^{-6}$). Readers put more work into each word and moved past fewer of them, while remaining exactly as sensitive to which words deserved the work.

2.6 Skimming and mind-wandering are different states

If our measures were simply insensitive, they should also fail when coupling really does change. They do not (Fig. 5).

In the independent sample, readers read the same kind of sentences either for comprehension or while searching for a specific semantic relation. The instructed search made reading shallower on every behavioral measure. Total reading time fell by 20%, and readers fixated 30% fewer words. Coupling fell with it. The frequency slope on gaze duration dropped to 62.3% of its value in normal reading ($p = 1.7 \times 10^{-4}$), and surprisal behaved identically (62.7%, $p = 7.4 \times 10^{-5}$).

The neural response did not follow. The occipital frequency kernel changed by +32.2% ([−30.6, +88.4]%, $p = 0.33$, $n = 12$), and attenuations greater than 20% are excluded, against a behavioral attenuation of 34 to

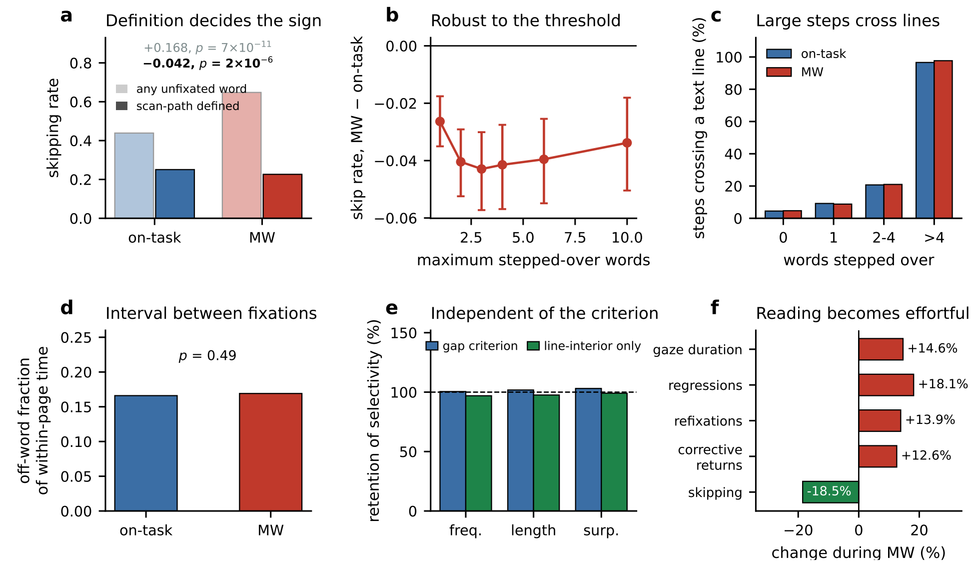

Figure 4: The skipping marker depends on the definition, and the corrected profile shows more effort rather than less. (a) Skipping rate under a naive definition (any unfixated word, pale) and a scan-path definition (solid). (b) The corrected effect across thresholds for the number of words a single forward saccade may step over; bars are 95% bootstrap intervals. (c) Percentage of forward steps that cross a text line, by the number of words stepped over. Steps larger than four words are return sweeps in both states. (d) Fraction of within-page time not spent on a mapped word. (e) Retention of skipping selectivity under the gap criterion and under a purely layout-based criterion that uses only line-interior words. (f) Change in each eye-movement measure during mind-wandering, computed within reader and averaged.

44%. With twelve readers this is a weak constraint, but it is a constraint, and it points the same way as the frequency result in the primary sample.

This gives a three-way contrast that we think is the most informative comparison in the paper. Under an instructed shallow goal, the brain continues to register each fixated word while the eye-movement policy stops tracking word properties. Under spontaneous mind-wandering, neither does. Skimming and mind-wandering are therefore not two points on a single axis of disengagement. They are different states with different signatures. Skimming changes what the eyes are being asked to do. Mind-wandering leaves that intact and changes something else.

Task and recording session are partly confounded in that dataset, because task order was fixed for every reader. A third task split across both sessions separates the two influences: there is a genuine session effect (frequency coupling at 84.9%, $p = 0.020$), but the task effect survives within a single session, at 66.4% for frequency ($p = 0.002$) and 56.0% for surprisal ($p = 1.3 \times 10^{-4}$). Two further caveats, a predeclared lexical-versus-semantic dissociation that was not supported and a scale artefact that understates both of our effects on the conventional log scale, are reported in SI S5 and SI S6.

2.7 What changes is global state, not word-level processing

Mind-wandering is not invisible to these measures. What it changes is not specific to any word. Four results establish that, and are reported in full in the Supplementary Information.

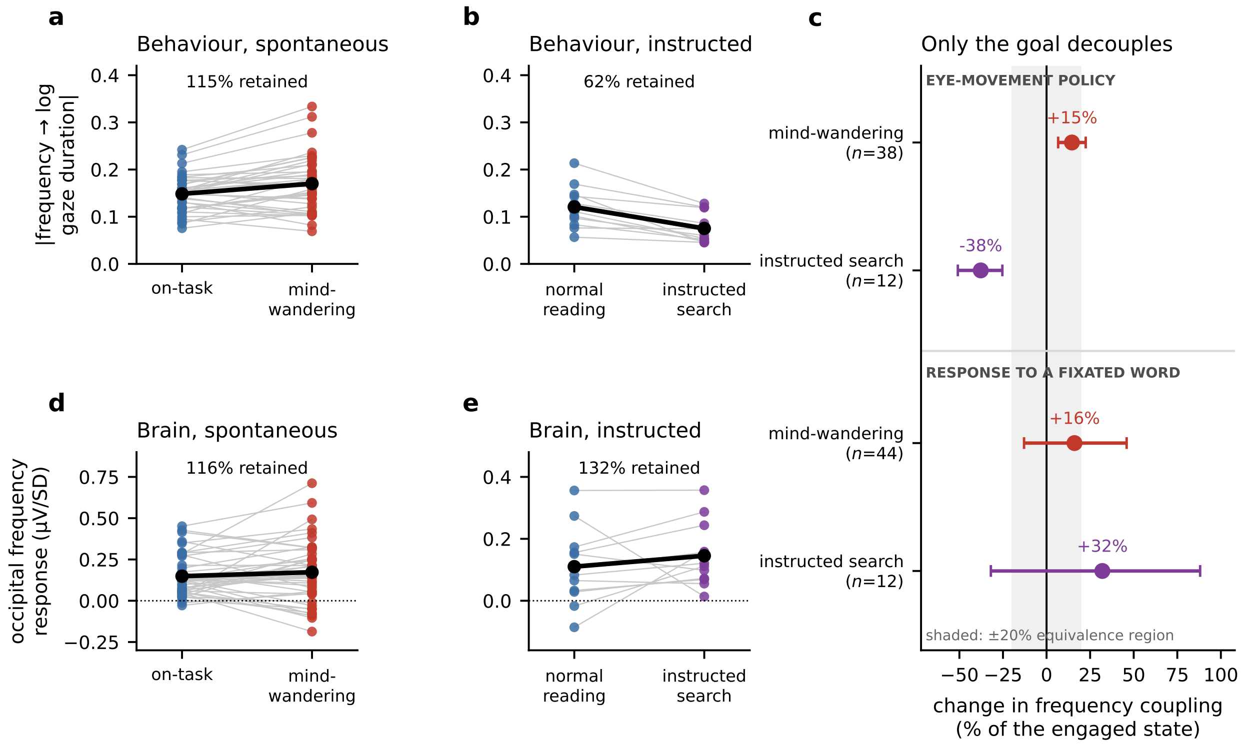

Figure 5: Skimming and mind-wandering are different states. (a, b) Frequency slope on log gaze duration, one point per reader, in the primary sample (on-task versus mind-wandering) and in the independent sample (normal reading versus instructed relation search). Black lines join condition means. (d, e) The occipitotemporal frequency response over 150 to 290 ms in the same four conditions, one point per reader. (c) All four cells on one axis, as the change in coupling expressed as a percentage of the engaged state, with 95% bootstrap intervals over readers. The point estimate is the mean per-reader change divided by the group engaged-state effect, so a negative value is an attenuation. Shading marks the ±20% equivalence region. The instructed goal attenuates the eye-movement policy and leaves the brain response alone; mind-wandering does neither. The neural interval in the instructed sample is wide, and excludes only attenuations greater than about 20%.

First, the fixation-related response does change in size, and only in size. Over 150 to 290 ms the mind-wandering minus on-task field is reliable and is strongly anticorrelated with the ordinary fixation field ($r = -0.872$): it is a scaled copy of it. A single proportional factor of 25.8%, fitted leave-one-reader-out, accounts for 76.1% of the group map and leaves no reliable residual ($p = 0.277$). A change of that kind applies equally to every word, so it cannot carry information about which word was processed less, in the way a change in the frequency or surprisal kernel would (SI S1, Fig. S1).

Second, the extra effort is not aimed anywhere. The mind-wandering increase in fixation duration, regressions and refixations was flat across quartiles of word difficulty (trends $p \geq 0.25$), and the contribution of extended context to reading time and to the N400 was unchanged ($p = 0.37$ and $p = 0.28$). Readers are not selectively going back over the words that gave them trouble. Only regressions accumulate: the rate climbs steadily through an episode (+0.022 per additional two seconds, $p = 0.0012$), which 300 matched on-task stretches never reproduce (SI S2).

Third, slow state markers move. Pupil diameter rose (+0.20 z, $p = 1.5 \times 10^{-4}$) and central beta power fell (-0.074 z, $p = 2.4 \times 10^{-4}$) at the onset of a reported span, neither at position-matched control onsets. Readers also became less like one another, but only in the component that word properties do not explain: overall inter-subject alignment barely moved (97.1% retention, $p = 0.30$), while alignment in the residual after regressing out frequency, length and surprisal fell to 86.7% ($p = 0.037$, permutation $p = 0.007$) (SI S3, Fig. S2).

Fourth, a continuous index of how strongly the text is currently driving a reader, built without any reference to the reports, does not detect mind-wandering (AUC 0.506, $p = 0.18$), while an index of how hard the reader is working does (0.528, $p < 10^{-4}$). Detectors that appear to measure attention to text are measuring effort and arousal (SI S4, Fig. S3).

2.8 Comprehension failure tracks the location of the lapse, not of the gaze

Word processing is preserved, but comprehension is not. This is the second half of the dissociation, and it is where the consequence of a lapse becomes visible. The released dataset includes one multiple-choice question per page, which gives an outcome that is independent of the physiology. It provides the first test of the reports against something other than another measure of the reader's body (Fig. 6).

At page level the reports behave as they should. Accuracy fell from 0.673 to 0.552 on pages with a reported span. In the tightest specification, which holds reader and story constant, the cost is 5.9 accuracy points ($p = 0.006$), a within-cell permutation gives $p = 0.014$, and the effect is graded with the amount of the page covered ($p = 7.2 \times 10^{-4}$). It is specific to the page whose question is asked rather than to the rest of the story.

The more informative question is where on the page it matters. We annotated each item with its answer-bearing sentences, a median of 21% of the page, and validated the annotation independently. A language model scoring from those sentences alone recovers the gold answer at 0.859, against 0.848 from the whole page. The spans really do carry the answers.

Mind-wandering on the answer span cost -0.073 in accuracy (95% CI [-0.092, -0.053], $p = 1.7 \times 10^{-13}$), with reader and item held constant. Mind-wandering elsewhere on the same page cost -0.035 and was not reliable ($p = 0.064$). Accuracy runs 0.680 with no lapse anywhere, 0.640 with a lapse on the page but not on the answer, and 0.405 with a lapse on the answer itself.

Two controls carry this result. First, we repeated the analysis on 1000 randomly placed regions of the same size on the same page. The observed effect is more extreme than all 1000, and a size-matched control region gives essentially nothing (-0.0017). Second, and more useful, the effect across those random regions scales linearly with how much each one overlaps the true answer span (slope -0.072, $p = 1.5 \times 10^{-14}$), and falls to -0.004 at zero overlap. Region size is held constant, so the only thing varying is location. Damage tracks distance from the answer. The effect also survives adding coverage, fixation count and dwell time on the span (-0.053, $p = 4.6 \times 10^{-6}$), so it is not simply that those readers did not read it.

That a reader misses what they were not attending to is not surprising on its own. The informative part is that the same reasoning fails when applied to the eyes. Reading the answer span more does predict accuracy at first glance (+0.033, $p = 0.016$). A size-matched control region predicts it just as well (+0.025, difference $p = 0.65$). Against the random-region distribution the true answer span sits at the 38th percentile ($p = 0.62$).

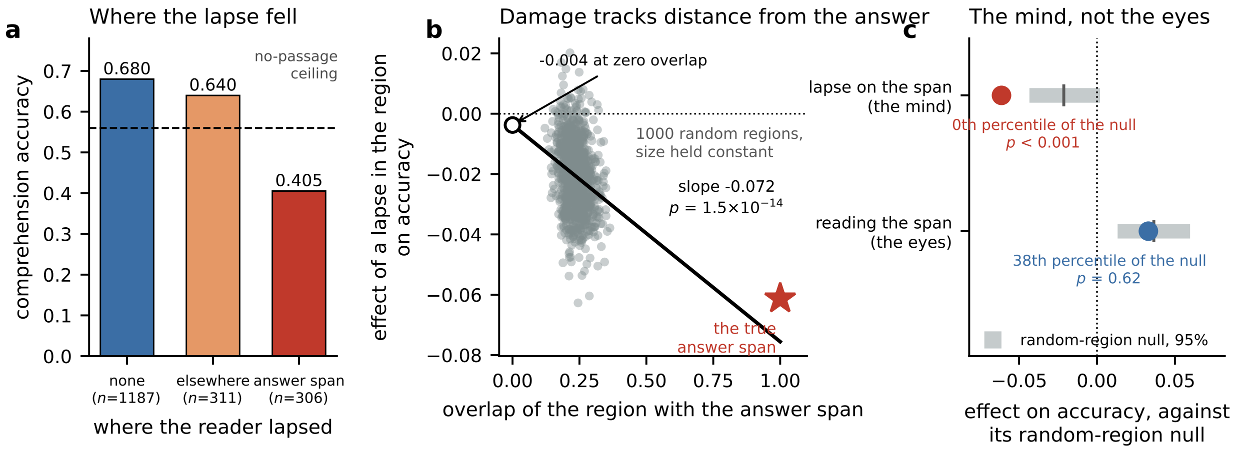

Figure 6: Comprehension failure tracks the location of the lapse, not of the gaze. (a) Raw accuracy by where the reported lapse fell, with the no-passage answerability ceiling marked. These are unadjusted proportions; the modeled effect holding reader and item constant is smaller. (b) The effect of a lapse on accuracy for each of 1000 randomly placed regions of the same size on the same page, against how much each region overlaps the annotated answer span. Region size is held constant, so only location varies. The open circle is the fitted value at zero overlap and the star is the true answer span. (c) The same comparison run twice. Where the mind was, at which specific words, is more extreme than all 1000 random regions. Where the eyes went sits inside the null. Bars are the random-region null intervals: the mind's from the stored 2.5th, 50th and 97.5th percentiles of its 1000 regions, the eyes' as mean ±1.96 standard deviations, which is the only summary of that null that was retained.

An independent set of span annotations reproduced this null at the 39th percentile. Where the mind was, at which specific words, is decisive. Where the eyes went is unremarkable.

The damage is also unflagged. A lapse on the answer span produced wrong answers (-0.061, $p = 4\times10^{-8}$) rather than expressions of uncertainty, which rose only marginally ($+0.021$, $p = 0.099$). Readers do not appear to know what they have lost.

We report the size of this effect in proportion. Adjusting for reader, item, reading amount and mind-wandering elsewhere, eliminating lapses on answer spans would raise accuracy by 1.79 points (95% CI $[0.80, 2.85]$), which is 4.9% of all errors (95% CI $[2.2, 7.7]$). The effect is large where it lands and modest in aggregate, because only 17.5% of trials carry a lapse on the answer span. The raw contrast between 0.680 and 0.405 overstates the population effect considerably.

This section carries a caveat that we would rather state than bury. The item bank was designed to encourage attentive reading rather than to measure it, and a language model with no passage at all answers 56% of the items correctly. The reading-dependent range is therefore narrower than the raw accuracies suggest. Because the localization effect is identified within item, item answerability is absorbed and cannot drive it, but the absolute sizes should be read with the ceiling in mind.

3 Discussion

The reader whose mind has wandered is still reading the words. What has stopped is the making of anything out of them. Across three levels of eye-movement control we could not detect any reduction in word sensitivity during self-reported mind-wandering, and the same held for the lexical component of the fixation-locked brain response. The strongest of these nulls is the one that matters most for the decoupling account. Whether a word is skipped is decided on parafoveal information before the word is fixated, and it is the decision the meta-analytic literature identifies as the most consistent behavioral marker of mindless reading (Mézière et al., 2026). In our data the lexical control of that decision was statistically equivalent to its on-task value within ±10%, several times tighter than the attenuations the decoupling account predicts and than the one our own instructed-goal condition produced.

We take the positive control seriously. An instructed shallow goal attenuated the same frequency slope by a third within a single session. Readers who are told to search a sentence for a specific relation stop letting word difficulty govern how long they look. So the measures are capable of registering a change in coupling, and their failure to register one during mind-wandering is evidence rather than absence of evidence.

Why, then, has the field converged on the opposite conclusion? Part of the answer is likely the measurement issue we document. In continuous multi-page reading, scoring any unfixated word as skipped conflates a saccade that steps over a word with a gap in the correspondence between gaze and text. The two are not equally distributed across states, and the resulting bias is large enough to invert the sign of the effect. We do not claim that this artefact is present in prior work, and we are not in a position to know. The operational definition of skipping is frequently not reported in enough detail to tell. Much of the contributing literature also used isolated sentences, where the scan path is short and gaze is easier to keep in register with the text. The point is narrower and, we think, more useful. The marker is definition-sensitive in continuous reading, the two definitions can differ in sign, and the diagnostic that separates them is cheap to run.

A second part of the answer may be paradigm. Reichle et al. (2010) used self-caught reports during sentence-by-sentence reading of a novel, and analyzed the intervals leading up to a report. Our spans are retrospective, and their onsets are correspondingly imprecise. It is possible that a brief decoupled interval exists immediately before a reader notices the lapse, and that our labels smear across it. That is a real limitation, but it cuts both ways. It means our estimate is a property of sustained mind-wandering as readers themselves demarcate it, which is the state that matters for the everyday experience the phenomenon is named after.

The positive results point somewhere different from gating. Mind-wandering made reading more effortful. Readers looked longer, went back more often, refixated more often, and skipped less, all while remaining exactly as sensitive to which words merited that treatment. This is hard to reconcile with an account in which the perceptual stream is attenuated.

It is also hard to reconcile with the obvious alternative, that readers are repairing words they failed to process. Repair should be aimed at difficult words, and none of the three effort measures was. A direct test of whether extended context still contributes to reading time and to the N400 was likewise null. Readers are not selectively going back over the words that gave them trouble. Nor is all of the extra effort the same kind of thing. Longer fixations and refixations appear immediately at the start of an episode and stay flat, which is what a change of state looks like. Regressions instead climb steadily for as long as the episode lasts, and do so specifically during reported spans. Something accumulates, it expresses itself as undirected backtracking, and it is indifferent to which words are hard.

Two positive results tell us where to look for the failure. The first is the loss of alignment across readers. Word-level coupling is low-dimensional. A reader can remain perfectly sensitive to frequency and surprisal while constructing something quite different from the text, and the inter-subject measure is sensitive to that residual in a way that word-level regression is not (Simony et al., 2016; Nastase et al., 2019). The second is the comprehension result. Mind-wandering predicts failure on the question for that page, and the cost is concentrated on the specific sentences that carry the answer rather than spread across the page. Both point the same way. What fails is not the identification of words but what is built from them.

The comprehension analysis also sharpens the main claim in a way the coupling results alone cannot. If word processing is intact, one might expect gaze on the answer-bearing sentences to predict whether the answer was retrieved. It does not, once a size-matched control region and a random-region distribution are taken into account. The subjective report of where attention was carries content-specific information that the objective record of where the eyes went does not. That asymmetry is difficult to reconcile with an account in which mind-wandering works by withdrawing the eyes from the words.

Taken with the finding that inter-reader alignment falls only in the component that word properties do not explain, the account we would put forward is that the reader loses the thread rather than the words. Word identification continues and the eyes keep responding to it. What degrades is the higher-order representation that makes one word follow from the last. Its behavioral expression is a reader who goes back more and more as the episode runs on, without knowing quite where to go back to. We stress that this is the account our data are consistent with rather than one they establish. What the data localize is the level at which reading fails: downstream of word identification and of the eye-movement policies that word identification drives. We cannot say which downstream process fails. Integration into a situation model, its maintenance in working memory, and the selection that keeps it updated are all candidates, and this design does not separate them. The two measures that would test it directly, the N400 and the contribution of extended context, are respectively unmeasurable here and null. The state changes we observed, a pupil dilation and a cortical desynchronization time-locked to onset and absent at matched control onsets, are consistent with that reading, though neither is diagnostic of a specific mechanism on its own (Smallwood et al., 2011; Kam et al., 2011; Braboszcz and Delorme, 2011).

The rescaling result offers an account of one of the more puzzling features of this literature. Mind-wandering has a real neural signature, and yet fixation-locked activity carries very little information about what a reader took from the text. A proportional reduction in the response to every word explains both facts at once. It is detectable, because the overall amplitude drops. It is uninformative about content, because it drops equally for every word. A change in the frequency or surprisal kernel would have been informative in that way, and there is no such change.

The neural side of this study is asymmetric and we would not want that glossed over. The frequency response is measured well enough to constrain the question: an attenuation of more than 8% is excluded, which is tighter than the effect an instructed goal produces. The surprisal response is not. Its amplitude is small enough relative to trial noise that 44 readers and 400,000 fixations cannot detect even its complete abolition, so the semantic channel remains open. Closing it will need either a paradigm that produces a larger N400, such as strongly constraining sentence contexts, or a great deal more data per reader. We checked whether the limit is instead the analysis. Aggregating evidence over all 64 sensors and the whole 0 to 500 ms window, with a text kernel fitted across readers and evaluated on a reader and an article it had never seen, yields a model that predicts held-out electroencephalography well ($p = 1.6 \times 10^{-7}$ occipitotemporally) and recovers the additive mind-wandering offset, yet resolves the surprisal channel no better than the single-window contrast does: 127% of the on-task effect detectable rather than 144%. Substituting language model representations for the word properties does not help either (Section S11). The limit is the signal, not the model. We say this plainly because a reader skimming the figures could otherwise take the two neural rows to carry equal weight, and they do not.

Two further limitations deserve statement. The mind-wandering and instructed-goal comparisons come from different samples, so the contrast between them generalizes across laboratories but does not identify the goal as the causal switch. And the corrected skipping analysis conditions on the reader having scanned the local region. That is a state-dependent selection, and although the conclusion survived a layout-based rule and an exclusion-matched analysis, no version of this measurement is free of the choice entirely.

What is established and what is inferred. The levels of this argument are not equally supported, and the vocabulary this literature uses for them is loose enough that they can be run together. We separate them explicitly. Word identification and the sensitivity of eye-movement policy to word properties are preserved, and preserved with stated bounds. Those bounds are ±10% on selection and 8% on the occipitotemporal frequency response, and an instructed goal produces a larger attenuation than either admits. That is the established part. Contextual integration is unresolved in either direction; the N400 test here is uninformative, and the behavioral evidence bearing on it is indirect. Global state clearly changes: pupil dilation, cortical desynchronization, and a proportional reduction of the fixation response. Comprehension clearly fails, and fails on the specific content the reader was absent for. The step from those facts to a failure in constructing and maintaining a passage-level representation is an inference we think is the most plausible available, not a measurement. Nothing here shows that attention never decouples from perception, and neural work outside reading gives good reason to think it sometimes does. What these data exclude is narrower and more useful. In the specific, testable form that predicts weakened lexical control of the eyes and of the fixation-locked response, the decoupling account does not describe sustained, self-demarcated mind-wandering in continuous reading. The failure lies downstream of word identification.

For applied work the implication is direct. Some systems infer attentional state from how the eyes respond to text, in tutoring software, in reading assessment, or in driver monitoring. Our data suggest that this signal is not diagnostic of spontaneous mind-wandering, whatever else it may track. In this dataset it tracked task set. A reader who has been told to skim and a reader whose mind has wandered produce opposite changes in word-level coupling, and confusing them will produce detectors that work in the laboratory and fail in the field.

The continuous index makes the same point in a form that is easy to test elsewhere. An index of how strongly the text is currently driving a reader, built without reference to any report, is at chance for mind-wandering, while an index of how hard the reader is currently working is not. Detectors that appear to measure attention to text are measuring effort and arousal. That may still be useful, but it is a different quantity, and the distinction matters as soon as a system is asked to decide whether a reader has understood something rather than whether they are tired.

4 Methods

4.1 Datasets

Primary sample. We analyzed the ROAMM dataset (Sun et al., 2026): 44 adults read five Wikipedia articles, each divided into ten pages of about 220 words, presented as monospaced text. Reading was self-paced and page-wise. After each article, readers marked the spans of text during which their minds had wandered, and answered one multiple-choice comprehension question per page. Electroencephalography was recorded with a 64-channel BioSemi ActiveTwo system at 256 Hz and binocular eye movements with an EyeLink 1000 Plus. The released derivative provides fixation events mapped to corpus word identifiers. We used all 404,557 first-pass fixations; 7.2% fell inside a reported mind-wandering span. The Comprehension scores are used in Section 2.8.

Positive-control sample. We used ZuCo 1.0 (Hollenstein et al., 2018a): 12 adults read isolated sentences under three tasks, with 105 usable channels of EEG at 500 Hz and simultaneous eye tracking. Task 2 is normal reading of encyclopedic sentences, task 3 is a search for a named semantic relation in encyclopedic sentences, and task 1 is sentiment reading of film-review sentences. Task order was fixed for every reader: session one contained task 2 followed by the first half of task 1, and session two contained task 3 followed by the second half of task 1. We exploit that structure to separate task from session, splitting task 1 at its sentence-index midpoint.

4.2 Linguistic predictors

Word frequency was the Zipf value from a large subtitle- and web-derived corpus (van Heuven et al., 2014; Brysbaert et al., 2018). Contextual surprisal was the negative log probability of each word under GPT-2 (Radford et al., 2019), conditioned on the preceding text within the article and with the sentence-initial token conditioned on a beginning-of-sequence marker. Sub-word tokens were mapped to words by maximum character overlap; a containment rule undercounts words because of the model's leading-space convention. Word length was in characters.

4.3 Defining the three eye-movement decisions

Reading positions were assigned in corpus order within each article. For each reader and article we took the ordered sequence of first-pass fixation positions.

Selection. For each forward step from position p_i to $p_{i+1} > p_i$, the words strictly between were treated as stepped over. A word was scored as skipped if it was stepped over in a step spanning at most G intervening words and never fixated, and as fixated otherwise. Words falling only inside steps larger than G were excluded from the analysis set, since the reader is not demonstrated to have scanned them. The primary analysis used $G = 4$; results are reported for $G \in \{1, 2, 3, 4, 6, 10\}$. The naive alternative, in which any word in the reading span without a first-pass fixation counts as skipped, corresponds to $G = \infty$. Mind-wandering state for a stepped-over word was taken from the bounding fixations, and steps whose bounding fixations disagreed were excluded (3.4% of steps).

We also used a criterion that makes no reference to gap size. Word bounding-box tops were clustered into text lines within each page, with a tolerance of 40 pixels against a line spacing of about 90 pixels, because tops jitter by a few pixels with glyph height. A word was analyzable if the reader fixated at least one word on the same line both before and after it. This yields the same conclusions and is reported alongside the gap criterion.

Only fixations that the dataset providers could map to a word are present in the released derivative. We can therefore measure the interval between successive mapped fixations, but we cannot observe a fixation that fell outside every word bounding box. Our claim about off-text gaze is an inference from elapsed time: a fixation we cannot see would still occupy time, and the intervals we observe are too short to contain one.

Duration. Gaze duration was the sum of first-pass fixations on a word; first-fixation duration was the first of them. Analyses used the natural logarithm, with raw milliseconds reported alongside where the scale matters.

Repair. A corrective return was defined for every forward step that stepped over exactly one word: the event was coded 1 if any of the next five fixations landed on that word. We also computed the overall rate of backward steps and the rate of immediate refixation.

4.4 Coupling measures and inference

For the two binary decisions we used Somers' D, computed as $2A - 1$ where A is the probability that a randomly chosen positive case has a higher predictor value than a randomly chosen negative case (Somers, 1962). It is invariant to monotone transformations of the predictor and to the base rate of the outcome, which matters here because the base rates differ between states. For duration we used ordinary least-squares slopes on standardized predictors, computed within reader and within state.

Reader is the unit of inference throughout. We report the mean, a 95% percentile interval from 10,000 reader-level bootstrap resamples, a paired t test, and the number of readers showing the effect. Families of three tests were corrected by the Holm procedure (Holm, 1979). Retention is the mind-wandering estimate divided by the on-task estimate, always computed on signed rather than absolute values, since ratios of absolute values are inflated when the underlying slopes are noisy.

Nulls were supported rather than merely reported. We ran two one-sided tests against bounds of 10% and 20% of the on-task effect (Lakens, 2017) and computed default Bayes factors for the paired difference with a Cauchy prior of scale $1/\sqrt{2}$ (Rouder et al., 2009).

4.5 Within-token fixed-effects model

Because the same corpus token is read by many readers, mind-wandering status varies within token. We fitted

$$\log(\text{gaze duration})_{ij} = \alpha_i + \gamma_j + \delta\, m_{ij} + \sum_k \theta_k\, m_{ij} z_{k,i} + \varepsilon_{ij},$$

where i indexes token instance, j indexes reader, m is the mind-wandering indicator and z_k are standardized word properties. Token and reader effects were absorbed by alternating within-transformation, and standard errors were clustered on reader (Cameron and Miller, 2015). Main effects of word properties are absorbed by the token effects, which is the point: the interaction is identified purely from readers who differ in state on the same physical word in the same position of the same page.

4.6 Omnibus model-based coupling test

The unit is one fixation-to-fixation transition with both words mapped inside the page's genuine reading interval (392,766 transitions; the 0.9% whose target falls outside the candidate window are dropped). The candidate set for each transition is the words at page positions ±20 from the currently fixated word, which contains the true target on 99.1% of transitions.

Two heads are fitted. The target head scores every candidate and is trained with a softmax over the set; the duration head predicts log fixation duration. Each head receives a geometry block and, in the compared model, a text block. The geometry block holds page progress, line, position in line, distance to the line end, gaze position and its offset from the word center, incoming saccade amplitude, previous fixation duration, first-pass status, ordinal position in the run, and the number of times this reader has already fixated the candidate on this page. The text block holds Zipf frequency, length, within-sentence surprisal and the two extended-context terms, taken at the candidate and at the currently fixated word for the target head, and at the fixated word and $w-1$, $w+1$, $w+2$ for the duration head.

Both heads are trained on on-task transitions only. Evaluation is a 5×4 design of five article folds crossed with four reader groups, so each of the 20 fitted models is tested on transitions from an article and from readers it never saw. The target head is compared as two separately trained models and read out per transition as $\log_2 p_{\text{text}} - \log_2 p_{\text{no text}}$ on the word actually fixated. The duration head is two-stage: a geometry network is fitted first and the text block is fitted to its held-out residual, because the lexical contribution to duration is about 0.2% of the variance while the seed-to-seed variability of two independently trained networks exceeds it. Replacing the ridge text stage with a network made the duration read-out three times smaller (24 of 44 readers positive against 41), so the text-to-duration mapping is treated as linear.

Inference is at the reader level: per-reader means in each state, a 10,000-draw bootstrap over readers for intervals, and the same fixed-effect ladder and episode-preserving permutation used elsewhere in this paper. The permutation replaces each reader's mind-wandering labels with another reader's, transferred by article, page and word position, which preserves episode length, contiguity and where in the text episodes fall while breaking the link between a reader's own lapses and their own eye movements.

4.7 Control analyses

A relocation control tested whether the observed contrasts are specific to reported spans. For each reader we recorded the number and length of their mind-wandering spans, relocated each to a random on-task position within the same run, and recomputed the whole analysis. Two hundred relocations gave a null distribution for each contrast. We also repeated the primary analysis within tertiles of position through a run, restricted to spans marked as fully off-task, and leaving out each article in turn.

For the interpretation of large positional gaps we computed, for every consecutive fixation pair, the interval between the end of one fixation and the start of the next, and tabulated it against gap size, state, and whether the pair crossed a page boundary.

4.8 Electrophysiology

Fixation-related potentials were epoched from -100 to $+500$ ms around fixation onset. In natural reading successive fixations overlap, and the overlap is correlated with word properties because difficult words are fixated longer, so naive averages are not interpretable (Dimigen et al., 2011; Dimigen and Ehinger, 2021). We therefore estimated overlap-corrected kernels by linear deconvolution: a sparse finite-impulse-response design matrix was built over the epoch for an intercept and for each predictor, including the interactions with mind-wandering state, and solved by ridge-regularized normal equations for all channels at once (Smith and Kutas, 2015; Ehinger and Dimigen, 2019).

Regions of interest were fixed a priori: an occipitotemporal set analyzed from 150 to 290 ms for frequency, and a centroparietal set analyzed from 300 to 450 ms for surprisal. Beyond the regions of interest we ran cluster-based permutation tests over the full epoch and all channels with sign-flipping at the reader level (Maris and Oostenveld, 2007). Split-half reliability of the fixation-related potential was computed with trial counts matched between states.

4.9 Comprehension outcomes and answer spans

The released dataset records one four-option comprehension question per page. We parsed 2,200 per-page outcomes and validated them against the shipped accuracy fields and against the item keys. One item was found to be mis-keyed and is excluded everywhere. The mapping between the physiology subject index and the released subject identifiers was recovered from a 50-page reading-duration fingerprint rather than assumed, and is bijective.

Each item was localized to its answer-bearing sentences using sentence embeddings and inverse-document-frequency overlap with the question and its options, over the candidate sentences of that page. Seven items were hand-corrected where a fact is split across sentences. Items phrased with a negation were handled by taking the union of the sources of the true options. The annotation was validated by scoring a local language model on the evidence sentences alone and comparing with the full page.

Trial-level models are linear probability models absorbing reader and item, with standard errors clustered by reader and confidence intervals from a reader-level bootstrap. The random-region control draws 1000 contiguous regions of the same word count from the same page and repeats the analysis on each. The overlap gradient regresses the effect obtained for each random region on that region's word overlap with the true answer span. The attributable fraction is a counterfactual from the fitted model with the answer-span mind-wandering term set to zero, bootstrapped over readers.

4.10 Topographic rescaling analysis

To ask whether mind-wandering changes the pattern or the size of the fixation-evoked response we computed, per reader, overlap-corrected fixation responses for on-task and mind-wandering fixations matched on frequency, length, surprisal and fixation duration. The difference field was tested with a sign-flip test on global field power, and its spatial pattern was correlated with the on-task fixation field over the same interval, with a bootstrap over readers.

The proportional model estimates a single scalar α such that the difference field equals $-\alpha$ times the on-task field. It is fit leave-one-reader-out to avoid using a reader's own data in their own estimate. We report the variance of the group difference map that the rescaled on-task map accounts for, and we test the residual field after subtracting it with the same sign-flip procedure. Time-resolved differences use a familywise cluster correction across the epoch. Polarity is not retained when comparing topographies, because a polarity-preserving comparison confounds a change in sign with a change in configuration.

4.11 Index of text-driven reading

The index is fit only on on-task fixations and never sees a mind-wandering report. We regress log fixation duration on Zipf frequency, word length and contextual surprisal, with reader effects, over fixations of 50 to 1000 ms with a mapped word and finite predictors. This yields, for each fixation, a duration predicted from the text alone.

For each reader and run we then take a centered rolling window of 25 fixations and compute the slope of observed log duration on predicted log duration, with both quantities centered inside the window. Centering removes any additive change in reading speed, so the slope reflects only the degree to which word properties are currently governing behavior. Windows with insufficient variance in the predicted values are dropped. Slopes are winsorized at the 1st and 99th percentiles and converted to a within-reader percentile, giving a 0 to 100 score. The comparison index is the within-reader percentile of raw log fixation duration over the same fixations. State contrasts are computed per reader and tested with a reader-level bootstrap.

4.12 Episode time course and effort targeting

Episodes were the maximal runs of consecutive fixations carrying a mind-wandering label. For each fixation we recorded elapsed time since the first fixation of its episode and binned it into 0–2, 2–5, 5–10 and beyond 10 seconds. Each reader's values were expressed relative to their own on-task mean and a linear trend across bins was fitted per reader. The null came from 300 sets of matched on-task stretches drawn at random positions with lengths sampled from the observed episode-length distribution, analyzed identically. Time on task was controlled by regressing each reader's regression indicator on their position through the run before binning.

For the targeting analysis, words were split into quartiles of contextual surprisal within reader, and the mind-wandering minus on-task difference in each effort measure was computed per quartile, with a linear trend across quartiles fitted per reader. Extended-context integration was tested by decomposing GPT-2 surprisal into a within-sentence component and the additional reduction contributed by the preceding passage, then testing the interaction of the latter with mind-wandering.

4.13 State dynamics and inter-subject correlation

Peri-onset trajectories were computed over ± 8 s around the onset and offset of each reported span, z-scored within run, for pupil diameter and for band-limited power. Control onsets were drawn from on-task reading and matched on position within the run. Inter-subject correlation used a leave-one-reader-out group template of the response to each word. The template for a given reader was the mean over the other readers' *on-task* readings of that word, requiring at least five contributors, so that it represents a normative response rather than a mixture of states. The target reader's words were then correlated with the template separately for each state, subsampling on-task words to match the mind-wandering count. Three versions of the response were used: raw log gaze duration, the same z-scored within reader and state, and the residual after regressing out frequency, length and surprisal. Specificity was tested by permuting the mind-wandering labels within reader at matched count.

4.14 Software and availability

Analyses were run in Python with NumPy, SciPy, pandas, scikit-learn and PyTorch. Both datasets are public and neither was collected for this study. The primary dataset is ROAMM, OpenNeuro `ds007629` version 1.3.0 (Sun et al., 2026), released under CC0; the control dataset is ZuCo 1.0 (Hollenstein et al., 2018a), distributed through the Open Science Framework (Hollenstein et al., 2018b). Analysis code and the derived tables needed to reproduce every figure and table are available at the repository given in the code availability statement.

4.15 Preregistration status

The analyses of the selection and repair channels were specified in a written plan, with gates and directional predictions taken from the published literature, before the corresponding analyses were run; that plan is included with the code. They are nevertheless exploratory with respect to this dataset, which has been analyzed by us for other purposes. The comparison in the control sample was gate-declared. We report equivalence tests and Bayes factors for every null and regard preregistered replication as necessary.

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We thank the authors of the ROAMM and ZuCo datasets for collecting and openly releasing the recordings analysed here. Neither dataset was collected for this study and neither group of authors bears responsibility for the analyses or conclusions.

Data availability

Both datasets analysed here are publicly available and were collected by others. ROAMM is distributed by OpenNeuro as `ds007629` under a CC0 licence and this study used version 1.3.0 (`https://openneuro.org/datasets/ds007629/versions/1.3.0`); the version matters because word-to-fixation mapping changed between releases. ZuCo 1.0 is distributed through the Open Science Framework (`https://osf.io/2urht/`). The derived tables that the figures and tables of this paper are computed from are included with the analysis code, so every reported number can be checked without reprocessing the raw recordings.

Code availability

All analysis code is openly available at `https://github.com/anirudhmn/MindWandering` (version 1.0) under the MIT licence. The repository contains the preprocessing and deconvolution pipelines, the scripts that produce every figure and table in this paper and its supplementary information, the machine-readable result file behind each reported test, and the preregistration documents referred to in the preregistration statement. It also ships the derived tables the analyses read, so every reported number can be recomputed without the raw recordings; model fitting is seeded per reader, fold and repetition. The raw recordings are obtained separately from the sources given above.

Ethical statement

The analyses in this manuscript use two previously collected, publicly available datasets (Sun et al., 2026; Hollenstein et al., 2018a). No new data were collected. The original data collections were approved by the institutional review boards of the institutions that carried them out, and written informed consent was obtained from all participants prior to those experiments.

Conflict of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary Information

These analyses are reported in full here rather than in the main text because each supports the argument without being part of its spine. Sections S1 to S4 describe what does change during mind-wandering, summarized in Section 2.7. Sections S5 to S7 concern the control sample. Sections S8 and S9 record two reliable but peripheral observations.

S1. The fixation response rescales rather than reconfigures

If word processing is preserved, the brain response to a fixated word should still differ between states in some way, because fixation-related potential amplitude does shift. The question is what kind of difference it is. A response can change in two ways. Its pattern can change, meaning different sensors are involved or the timing differs, which would point to a different configuration of underlying sources. Or the same pattern can simply get smaller, which would point to the same processing at lower gain.

These predict different things about the difference map. If the pattern changes, the difference between states should have a shape of its own. If only the size changes, the difference should look like a scaled copy of the ordinary response (Fig. S1).

It looks like a scaled copy. Over 150 to 290 ms the mind-wandering minus on-task field is reliable (global field power 0.067 µV, sign-flip $p = 2.0 \times 10^{-4}$) and is strongly anticorrelated with the on-task fixation field itself ($r = -0.872$, bootstrap 95% CI $[-0.931, -0.605]$). Estimating a single proportional scaling factor with a leave-one-reader-out fit gives an attenuation of 25.8% (95% CI [13.8, 39.7]), which accounts for 76.1% of the variance in the group map. After subtracting that rescaling, no reliable residual field remains (0.032 µV, $p = 0.277$). The effect occupies 172 to 266 ms (familywise cluster $p = 0.0056$).

One number therefore describes the whole neural effect of mind-wandering on the response to a fixated word. The response keeps its shape and loses about a quarter of its size. An earlier analysis of these data, which retained polarity when comparing topographies, appeared to show a novel scalp configuration. That was a property of the test rather than of the data, and the rescaling analysis resolves it.

This is why the main text treats the neural change as a change of state. A proportional change in amplitude applies equally to every word. It cannot carry information about which particular word was processed less, in the way a change in the frequency or surprisal kernel would. The neural signature of mind-wandering is real, and it is uninformative about content by construction.

These are sensor-level results. We do not interpret the scalp map as source localization.

S2. The extra effort is undirected, and one part of it accumulates

If the extra looking were repair, it should be aimed at the words that need repairing. It is not. Splitting words into quartiles of contextual surprisal within each reader, the mind-wandering increase was flat across quartiles for fixation duration (trend $p = 0.25$), regressions ($p = 0.33$) and refixations ($p = 0.54$). The same holds for the one direct test of extended-context integration we can run: comparing surprisal computed over the whole preceding passage against surprisal computed within the sentence alone, the contribution of extended context to reading time and to the N400 was unchanged during mind-wandering ($p = 0.37$ and

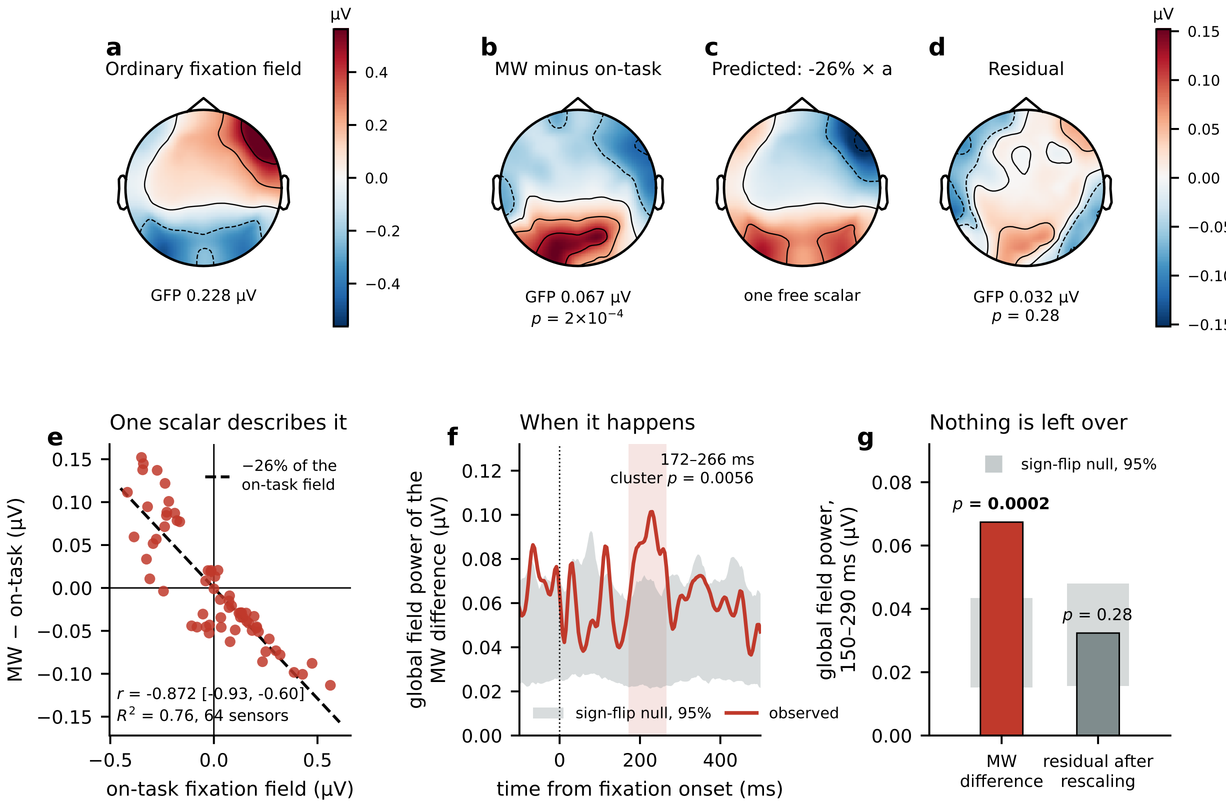

Figure S1: Mind-wandering rescales the fixation response rather than reconfiguring it. (a) The overlap-corrected response to a fixated word on task, averaged over 150 to 290 ms, as a scalp field. (b) The mind-wandering minus on-task difference field over the same window, on its own color scale. (c) What a single proportional scaling of panel a predicts, using the leave-one-reader-out factor of −25.8%. One free parameter. (d) What is left after subtracting that prediction. (e) The same relationship channel by channel. Each point is one of the 64 sensors; the dashed line is the fitted proportional model. (f) Global field power of the difference field through the epoch, against a sign-flip null over readers. Shading marks the familywise-corrected cluster. (g) Global field power in the analysis window before and after the rescaling, each against its own sign-flip null. Panels are sensor-level and are not interpreted as source localization.

$p = 0.28$). Whatever is failing, it is not showing up as a selective loss of context-based prediction. The strongest available version of this test is model-based. Taking the residual left after the geometry model and the on-task text model of Section S10, and asking whether any word property predicts the additional dwell during mind-wandering, cross-validated by reader, the held-out R^2 is −0.005 on mind-wandering transitions ($n = 28{,}450$) and −0.003 on on-task transitions ($n = 364{,}316$). Nothing a model can read from the text explains where the extra looking goes.

The three effort measures do, however, behave differently over time. Fixation duration and refixation are step changes: fully present in the first two seconds of an episode and flat thereafter (trends $p = 0.63$ and $p = 0.49$; against matched on-task stretches, $p_{\text{perm}} = 0.52$ and 0.43). Regressions are not. The regression rate rises steadily through an episode, from +0.016 above the reader's own baseline in the first two seconds to +0.080 after ten seconds (linear trend +0.022, 95% CI [0.011, 0.034], $p = 0.0012$, 26 of 37 readers). This is specific to reported spans: 300 matched on-task stretches of the same length distribution never produced a trend this large ($p_{\text{perm}} < 0.005$), and it survives removing each reader's position through the run (+0.022, $p = 0.0007$), so it is not time on task.

Something therefore accumulates across a mind-wandering episode, and what it produces is backtracking rather than slower or more repetitive fixating. It does not accumulate in a way that tracks word difficulty.

S3. Slow state changes and inter-subject alignment

Two things changed reliably (Fig. S2a–c). Time-locked to the onset of a reported span, pupil diameter increased (+0.20 z, $p = 1.5 \times 10^{-4}$) and central beta power decreased (−0.074 z, $p = 2.4 \times 10^{-4}$), neither of which appeared at position-matched control onsets drawn from on-task reading. This is a slow change in global state, and it is the natural counterpart of the additive shifts in fixation duration and fixation-related potential amplitude reported in the main text.

Second, readers became less like each other, and it is worth being precise about which part of them. Using a leave-one-reader-out template built from other readers' on-task gaze durations for the same word, overall alignment barely moved (retention 97.1%, $p = 0.30$), and removing the additive and scale shift by z-scoring within state changed nothing (97.0%, $p = 0.29$). Most of what readers share is driven by word properties, and that component is preserved, exactly as the coupling analyses in the main text would predict. Removing it is what exposes the effect. In the residual after regressing out frequency, length and surprisal, alignment fell from 0.201 to 0.175 (retention 86.7%, $\Delta = -0.027$, 95% CI $[-0.051, -0.002]$, $p = 0.037$), and shuffling the mind-wandering labels within reader at matched count never reproduced it ($p_{\text{perm}} = 0.007$). The earlier report of a drop in raw gaze inter-subject correlation in these data ($\Delta = -0.028$, $p = 0.025$) is consistent with this once the difference in template construction is taken into account.

So the loss of alignment is confined to the part of gaze behavior that word properties do not explain. Readers keep responding to the same words in the same way. What diverges is everything else.

S4. A continuous index of text-driven reading does not move

Every automatic measure of mind-wandering we are aware of is trained on mind-wandering reports. Asking whether such a measure captures attention is therefore circular, because it captures whatever the reports capture. We built an index that is never shown the reports, and asked what it does (Fig. S3).

The logic is the first principle the paper started from. If a reader is attending to the text, properties of the words should be governing behavior. We fit that relationship on on-task fixations only, predicting log fixation duration from frequency, length and surprisal with reader effects (Zipf −0.0128, length −0.0014, surprisal +0.0028). Every fixation then has a duration that the text alone predicts. In a rolling window of 25 fixations we regressed observed duration on predicted duration, with both centered inside the window. The centering is what makes the index specific. It removes any overall speeding or slowing, and leaves only the question of whether harder and rarer words are currently getting proportionally longer looks. We scored the result as a within-reader percentile, so the index runs from 0 to 100.

The index does not distinguish mind-wandering from on-task reading. Its mean is 50.1 on task and 48.4 during reported spans, a difference of −1.6 points (95% CI $[-4.3, +0.7]$, $p = 0.18$), with 24 of 44 readers in

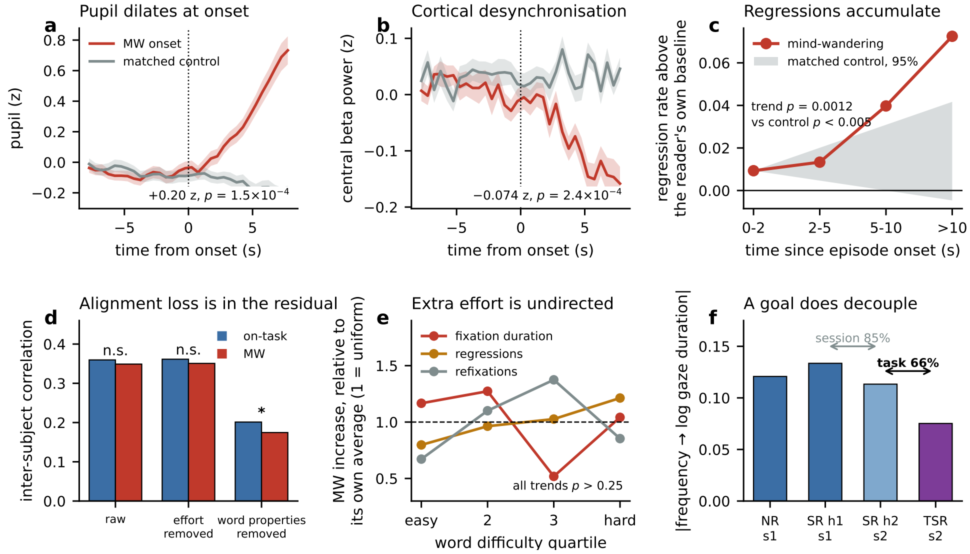

Figure S2: What changes during mind-wandering, and the positive control. (a, b) Pupil diameter and central beta power time-locked to the onset of a reported span, against position-matched control onsets. Shading is the standard error across readers. (c) Regression rate through an episode, relative to each reader's own on-task baseline. The band is the 95% range of trends from 300 matched on-task stretches of the same length distribution. (d) Inter-subject alignment against a leave-one-reader-out template, before and after removing the additive shift and the word-property component. (e) The mind-wandering increase in each effort measure across quartiles of word difficulty, scaled to its own average. A flat line means the increase is uniform. (f) Frequency coupling in the independent sample by condition, showing the session effect between the two halves of the sentiment task and the task effect within the second session.

the predicted direction, which is chance. As a classifier of individual fixations it reaches an area under the curve of 0.506.

The same pipeline applied to a level-based index behaves differently. Taking the within-reader percentile of raw fixation duration, that index rises from 49.8 on task to 52.8 during reported spans (+3.1 points, 95% CI [+2.1, +4.0], $p < 10^{-4}$, area under the curve 0.528).

The two indices differ in one respect. One tracks the slope relating word properties to behavior. The other tracks the level of behavior. Mind-wandering moves the level and leaves the slope alone. This is the same conclusion as the rest of the paper, expressed as a signal that runs continuously through a reading session rather than as a set of interaction terms.

S5. The predeclared lexical-versus-semantic dissociation

A predeclared test for a selective dissociation between lexical and semantic attenuation under the shallow goal was not supported, and we do not report one. The apparent effect was an artefact of collinearity between the two predictors, and the bivariate estimates attenuate about equally.

S6. A scale artefact in how coupling is reported

We also note a scale artefact that runs through this literature. When coupling is expressed as a slope on log duration, a purely additive change in duration rescales the slope by the change in the mean. Mind-wandering

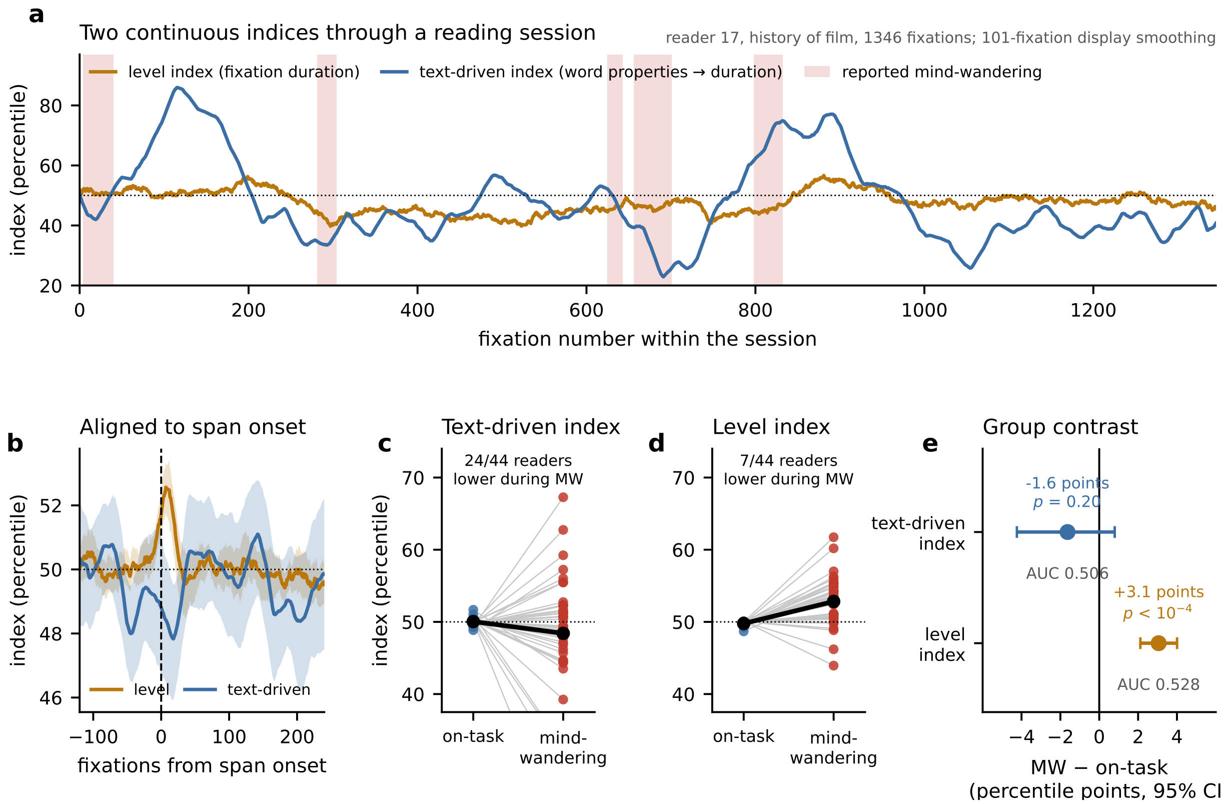

Figure S3: A continuous index of text-driven reading does not move; a level index does. (a) Both indices through one reader's session. The text-driven index is the rolling slope of observed on text-predicted log fixation duration, centered within the window; the level index is the within-reader percentile of raw fixation duration. Shading marks reported mind-wandering. The session is the one whose two state contrasts fall closest to the group means among sessions with at least five reported spans, and the traces are smoothed over 101 fixations for display only. (b) The same two indices aligned to the onset of a reported span, averaged within reader and then across readers, with 95% bootstrap bands. Descriptive; the tests are in (c) to (e). (c, d) Each reader's mean index by state. (e) The state contrast for both indices, with 95% bootstrap intervals over readers, and the area under the curve for classifying individual fixations. The index that never sees a mind-wandering report is at chance.

slows reading by 16%, so an unchanged coupling would appear as 86.2% retention on the log scale; instructed search speeds reading by 11%, so an unchanged coupling would appear as 112.0%. Both of our effects are therefore understated on the scale on which they are conventionally reported. On the raw millisecond scale, mind-wandering retention is 133.8% and instructed-search retention is 53.4%.

S7. Session and materials in the control sample

The contrast is also partly confounded with recording session, because the task order in that dataset was fixed for every reader and the two tasks fell in different sessions (Hollenstein et al., 2018a). A third task, sentiment reading, was split across both sessions, and splitting it at its midpoint separates the two influences. There is a genuine session effect: frequency coupling fell to 84.9% between the first and second half of the sentiment task ($p = 0.020$), with task and materials held constant. The task effect nevertheless survives within a single session, at 66.4% for frequency ($p = 0.002$) and 56.0% for surprisal ($p = 1.3 \times 10^{-4}$). The same decomposition shows that materials are not neutral: reading film-review sentences produced stronger coupling than reading encyclopedic sentences (110.5% for frequency, 134.2% for surprisal). A comparison of the two deep tasks pooled across sessions therefore averages a materials effect against a session effect of the opposite sign, and cannot be read as evidence that materials are irrelevant. The split assumes that sentence index corresponds to presentation order, which the dataset documentation supports but which we could not verify directly.

S8. Onset intervals

The fixation step at the start of a reported span was followed by an interval longer than 500 ms in 6.8% of cases, against 0.6% of steps taken within a state. The enrichment is elevenfold. Because span onsets are marked retrospectively, we cannot tell whether these long intervals precipitate the lapse, coincide with it, or are what makes a reader remember where the lapse began, and we draw no inference from them.

S9. Materials and coupling

In the control sample, frequency and surprisal coupling were stronger for film-review sentences than for encyclopedic sentences (110.5% and 134.2% respectively, within the first session). We report this because it bears on how the deep-reading conditions of that dataset should be compared, not because we can explain it.

S10. Robustness of the omnibus coupling test

The per-transition read-out is heavy-tailed, with a standard deviation of 0.68 bits against a mean of 0.138 and a median above the mean, and retention is a ratio of means. We therefore re-derived retention under eleven exclusions and two robust estimators. No refitting is involved; every variant reads the same frozen per-transition values.

Variant	Transitions	Retention	95% CI
All transitions	392,766	0.911	[0.822, 0.995]
Trimmed outside the 1st and 99th percentile	384,862	0.897	[0.813, 0.978]
Trimmed outside the 5th and 95th percentile	353,448	0.926	[0.852, 0.995]
Winsorized at the 1st and 99th percentile	392,766	0.908	[0.822, 0.990]
Winsorized at the 5th and 95th percentile	392,766	0.909	[0.828, 0.986]
Median rather than mean per reader	392,766	0.942	[0.878, 1.005]
Saccades of ten words or fewer	355,495	0.881	[0.785, 0.972]
Saccades of five words or fewer	347,252	0.911	[0.814, 1.005]
Fixations of 80 to 800 ms	382,711	0.902	[0.810, 0.990]
Episode interiors only	384,421	0.903	[0.815, 0.990]
Forward saccades only	249,929	0.965	[0.877, 1.056]

Retention never leaves 0.88 to 0.97. The most outlier-resistant estimator returns the smallest shortfall, so the tails contribute to it but do not produce it.

Retention appears higher for forward saccades (0.965) than for regressions and refixations (0.899), but that difference is in the denominator rather than the effect. In bits the shortfall is -0.0121 for forward saccades and -0.0110 for the rest, against on-task baselines of 0.163 and 0.108; the within-reader difference of differences is +0.0012 $[-0.026, +0.029]$, $p = 0.94$, with 20 of 40 readers in each direction. Forward saccades simply have more text-driven signal to lose. The shortfall is uniform across kinds of movement, which is the shape of a small global change rather than of one mechanism failing.

S11. A richer neural model does not rescue the surprisal response

We asked whether the uninformative surprisal contrast reflects the analysis rather than the data. The fixation kernels were refitted so that the text kernel is pooled across readers and every evaluated cell is held out on both the reader and the article, and the read-out became the improvement in held-out prediction of the continuous recording over the whole 0 to 500 ms window at all 64 sensors, rather than a coefficient in one window at one region.

The model works. It predicts held-out data occipitotemporally ($p = 1.6 \times 10^{-7}$, 36 of 44 readers) and centroparietally ($p = 6.3 \times 10^{-7}$, 37 of 44); permuting words within a page drives the same quantity below zero across two independent draws; and the mind-wandering kernel reproduces the additive occipitotemporal offset reported in the main text (+0.070 µV, $p = 7.8 \times 10^{-4}$).

It does not buy power. The smallest detectable change is 82% of the on-task effect for the frequency channel against 43% for the single-window contrast, and 127% against 144% for the surprisal channel. A held-out prediction gain is a noisier statistic than a directly estimated coefficient, and that loss outweighs what is gained by aggregating over sensors and latencies. Allowing the mapping from word properties to the kernel to be nonlinear changes nothing (+0.0006 μV^2, $p = 0.47$, 24 of 44 readers), which matches the behavioral result that a nonlinear text stage is worse than a linear one.

Replacing the word properties with GPT-2 layer representations does not help either. The best layer, the fourth, reaches a held-out gain of 0.0068 μV^2 against 0.0069 for the four word properties; prediction falls with depth, and the final layer falls below the word-permutation control. No layer detects a change smaller than twice the on-task effect, so the depth profile cannot address the question.

One caution follows from this analysis and applies beyond it. When the text block is restricted to word length, the held-out gain is larger at right lateral frontal sensors (0.0175 μV^2) than occipitotemporally (0.0075), the signature of residual ocular contamination, because length drives saccade amplitude. Removing length restores the occipitotemporal sensors to the top. Any predictor correlated with saccade amplitude can produce apparent coupling that survives amplitude nuisance regressors.